Eenheid 2.5

Note Title 2016/03/15

Limiete by oneindigheid en horisontale asimptote

Stewart pp 126 – 137

*Informele definisie van 'n limiet by oneindigheid p127

$\boxed{1}$ As f gedefinieer is op (a,∞), dan is $\lim_{x\to\infty} f(x) = L$ en waardes van $f(x)$ kom baie naby aan L as x positief groot genoeg geneem word

$\boxed{2}$ p 128

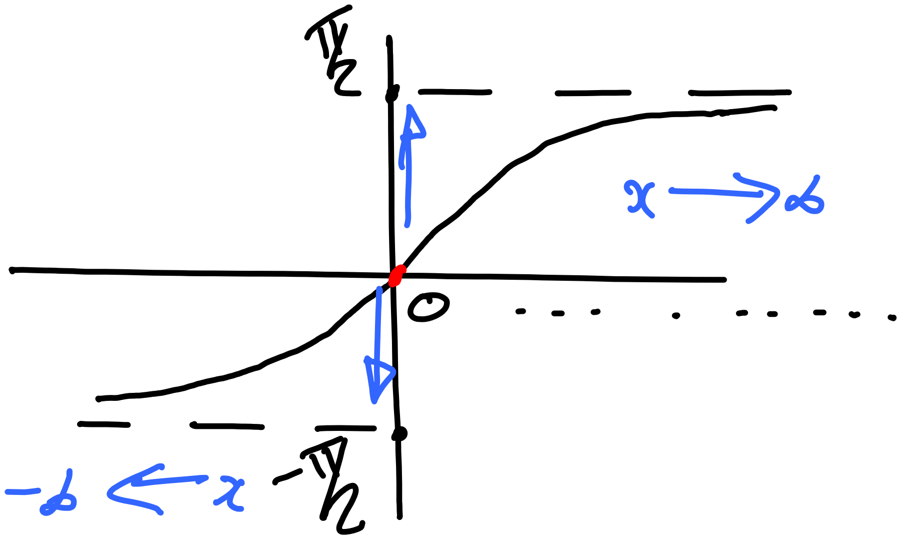
as f gedefinieer is op $(-\infty, a)$ dan is $\lim_{x \to -\infty} f(x) = L$ en waardes van $f(x)$ kom baie naby aan L as x negatief groot genoeg geneem word.

$\boxed{3}$ Die lyn $y = L$ is 'n horisontale asimptoot van 'n funksie as:

$\lim_{x \to \infty} f(x) = L = 0$ of $\lim_{x \to -\infty} f(x) = L = 0$

H.A. : $y = 0$ HA : $y = 0$

Voorbeeld : • $y = \text{bgtan}\, x$



$y = \text{bgtan}\, x$

$x > 0$: $\lim\limits_{x \to \infty} \text{bgtan}\, x = \pi/2$

HA is $y = \pi/2$

$x < 0$: $\lim\limits_{x \to -\infty} \text{bgtan}\, x = -\pi/2$

HA is $y = -\pi/2$

Dus $y = \text{bgtan}\, x$ is n kromme met Horisontale asimptote :

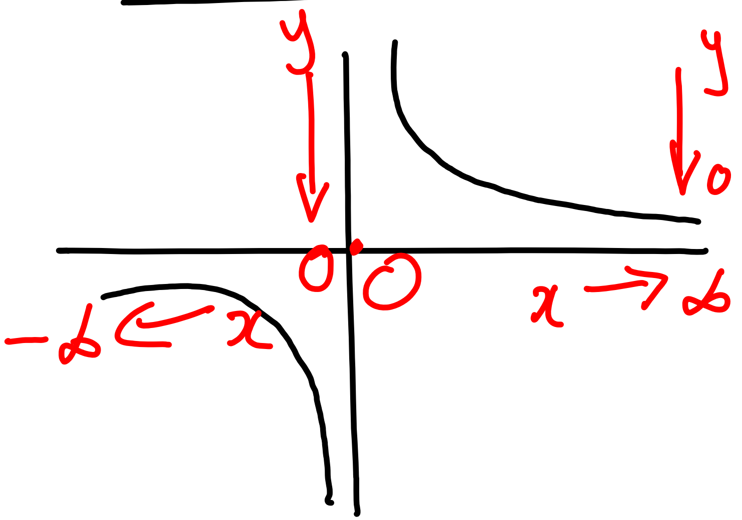
$x > 0$: HA is $y = \pi/2$ / $x < 0$: HA is $y = -\pi/2$

$\boxed{5}$ Stelling p129

As $r \in \mathbb{Q}$, $r > 0$ en x^r is gedefinieer vir alle x, dan is

$$\lim_{x \to \infty} \frac{1}{x^r} = 0 \quad \text{en} \quad \lim_{x \to -\infty} \frac{1}{x^r} = 0$$

Voorbeeld: $f(x) = \frac{1}{x^1}$ $(r = 1)$



$$\lim_{x \to \infty} \frac{1}{x^1} = 0$$

$$\lim_{x \to -\infty} \frac{1}{x^1} = 0$$

* **Rasionale funksie:** $f(x) = \frac{Q(x)}{P(x)}$

As jy 'n rasionale funksie het en jy bepaal $\lim_{x \to \infty} f(x)$ of $\lim_{x \to -\infty} f(x)$ dan hang die antwoord van f af van die graad van $Q(x)$ en $P(x)$.

Metode om HA ($y = L$) te voorspel

1) Graad van $Q(x)$ = Graad van $P(x)$

$4 = 4$

* Bv. $f(x) = \frac{2x^{4} - x^{3}}{4x^{4} + 2} = \frac{Q(x)}{P(x)}$

Graad is grootste mag v. x *

Dan is HA die volgende:

y gaan 'n Reële getal wees

2) Graad van $Q(x)$ $<$ Graad van $P(x)$ $\quad 3 < 4$

Bv. $f(x) = \dfrac{4x^3 - 3x^2 + 2x}{8x^4 + 2x^2 - 1}$

Dan is HA die volgende:

$y = 0$

3) Graad van $Q(x)$ $>$ Graad van $P(x)$ $\quad 3 > 2$

Bv. $f(x) = \dfrac{x^3 - x^2 + 7}{8x^2 + 7x}$

Dan is daar geen HA

- Bepaal HA van:

1) $f(x) = \frac{2x^4 - x^3}{4x^4 + 2}$

*Onthou: Altyd

Deel bo en onder deur GROOTSTE mag van x in NOEMER

Stelling

VOORSPEL:

HA is 'n reële getal

$\boxed{\lim_{x \to \pm\infty} \frac{1}{x^r} = 0}$

- $x > 0$:

$\lim_{x \to \infty} \frac{2x^4 - x^3}{4x^4 + 2}$

$= \lim_{x \to \infty} \frac{\frac{2\cancel{x^4}}{\cancel{x^4}} - \frac{x^3}{x^4}}{\frac{4\cancel{x^4}}{\cancel{x^4}} + \frac{2}{x^4}} = \lim_{x \to \infty} \frac{2 - \frac{1}{x^1} \to 0}{4 + \frac{2}{x^4} \to 0}$

$= \frac{1}{2}$

- $x < 0$

$\lim_{x \to -\infty} \frac{2x^4 - x^3}{4x^4 + 2} = \; ? \; = \frac{1}{2}$

Dus HA is $y = \frac{1}{2}$ vir $x > 0$ en vir $x < 0$

② $f(x) = \dfrac{4x^3 - 3x^2 + 2x}{8x^4 + 2x^2 - 1}$

Voorspel: HA is $y=0$

• $x < 0$: $\lim\limits_{x \to -\infty} \dfrac{4x^3 - 3x^2 + 2x}{8x^4 + 2x^2 - 1}$

$= \lim\limits_{x \to -\infty} \dfrac{\frac{4x^3}{x^4} - \frac{3x^2}{x^4} + \frac{2x}{x^4}}{\frac{8x^4}{x^4} + \frac{2x^2}{x^4} - \frac{1}{x^4}}$

$= \lim\limits_{x \to -\infty} \dfrac{\frac{4}{x} - \frac{3}{x^2} + \frac{2}{x^3}}{8 + \frac{2}{x^2} - \frac{1}{x^4}} = \dfrac{0}{8} = 0$

Net so as $x > 0$

$\therefore$ HA is $y=0$ as $x > 0$ en as $x < 0$

③ $f(x) = \frac{x^3 - x^2 + 7}{8x^2 + 7x}$

VOORSPEL : $3 > 2$
Geen HA

• $x > 0$: $\lim_{x \to \infty} \frac{\overset{\to \infty}{\boxed{x}} - 1 + \overset{\to 0}{\frac{7}{x^2}}}{8 + \overset{\to 0}{\frac{7}{x}}} = \infty$

Netso

• $x < 0$: $\lim_{x \to -\infty} f(x) = \infty$

$\therefore$ Geen HA nie

4 $f(x) = \dfrac{\sqrt{9x^2-1}}{4-8x} = \dfrac{\sqrt{x^2}\sqrt{9-\frac{1}{x^2}}}{x\left(\frac{4}{x}-8\right)}$ *

NB

$\boxed{\sqrt{x^2} = |x| = \begin{cases} x & \text{as } x>0 \\ -x & \text{as } x<0 \end{cases}}$

$x>0$: $\lim\limits_{x\to\infty} \dfrac{+\cancel{x}\sqrt{9-\frac{1}{x^2}}}{\cancel{x}\left(\frac{4}{x}-8\right)}$ ($\frac{1}{x^2}\to 0$, $\frac{4}{x}\to 0$)

$= \dfrac{\sqrt{9}}{-8} = -\dfrac{3}{8}$

$x<0$: $\lim\limits_{x\to-\infty} \dfrac{-\cancel{x}\sqrt{9-\frac{1}{x^2}}}{\cancel{x}\left(\frac{4}{x}-8\right)}$ ($\frac{1}{x^2}\to 0$, $\frac{4}{x}\to 0$) $= -\left(\dfrac{3}{-8}\right) = \dfrac{3}{8}$

HA : as $x>0$ $\quad y = -3/8$

as $x<0$ $\quad y = 3/8$

Grafieke:

5 $\quad f(x) = \text{bgtan}(-2x^3)$

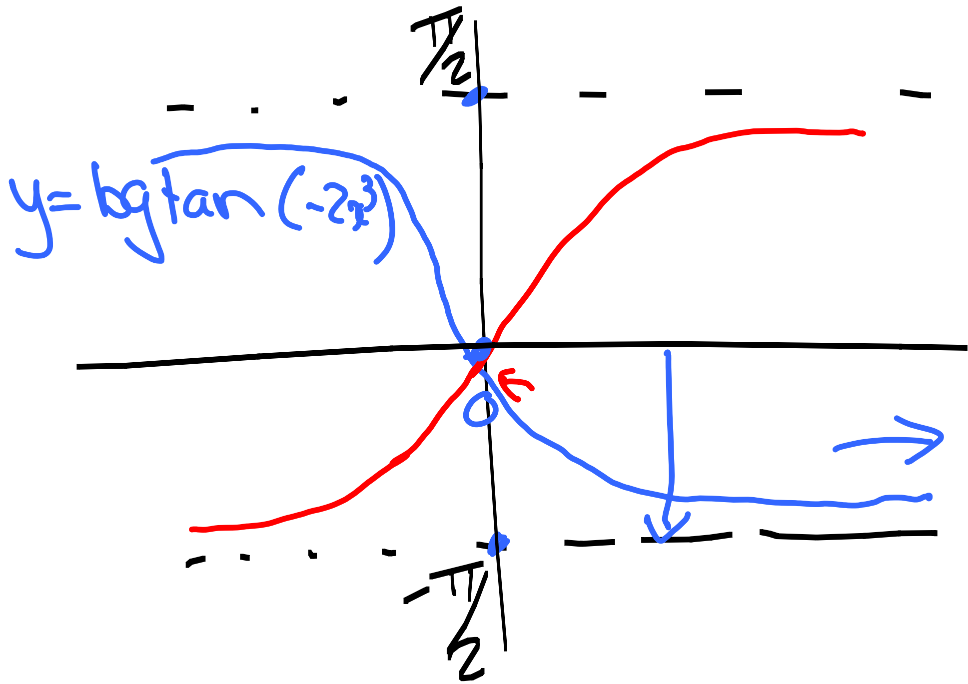
$y = \text{bgtan}\, x$

$x > 0$: HA is $y = -\pi/2$

want $\lim\limits_{x \to \infty} f(x) = -\pi/2$

$x < 0$: HA is $y = \pi/2$

want $\lim\limits_{x \to -\infty} f(x) = \pi/2$



$\pi/2 \approx 1.6$

$\pi \approx 3.14$

$y = \text{bg}\tan(2x^3)$

⑥ $f(x) = e^{3x}$

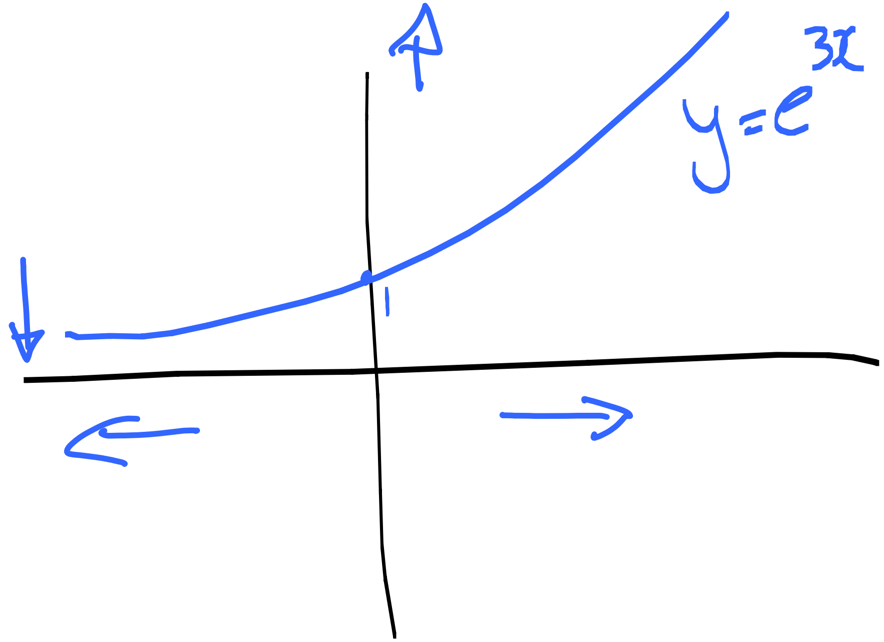
$x > 0$: HA bestaan nie

want $\lim_{x \to \infty} e^{3x} = \infty$

$x < 0$: HA is $y = 0$

want $\lim_{x \to -\infty} e^{3x} = 0$

Kyk na $f(x) = e^{-x}$!



Ekstra voorbeelde: HA ??

① $\lim_{x \to \infty} \frac{3x+5}{x-4} = \lim_{x \to \infty} \frac{\frac{3\cancel{x}}{\cancel{x}} + \frac{5}{x}}{\frac{\cancel{x}}{\cancel{x}} - \frac{4}{x}} = 3$

Stelling

$\frac{5}{x} \to 0$

$\frac{4}{x} \to 0$

$x > 0$: HA is $y = 3$

Net so as

$x < 0$: HA is $y = 3$

② $\lim\limits_{x \to -\infty} \dfrac{\sqrt{9x^6 - x}}{x^3 + 1}$

$= \lim\limits_{x \to -\infty} \dfrac{\sqrt{x^6}\sqrt{9 - \frac{x}{x^6}}}{x^3\left(1 + \frac{1}{x^3}\right)}$

$= \lim\limits_{x \to -\infty} \dfrac{\cancel{(-x^3)}\sqrt{9 - \frac{1}{x^5}}}{\cancel{x^3}\left(1 + \frac{1}{x^3}\right)}$

$\boxed{\begin{aligned} &\bullet \sqrt{x^6} = |x^3| \\ &= \begin{cases} +x^3 & \text{as } x > 0 \\ -x^3 & \text{as } x < 0 \end{cases} \end{aligned}}$

$\lim\limits_{x \to \infty} f(x)$

$= \lim\limits_{x \to -\infty} \dfrac{-\sqrt{9 - \frac{1}{x^5}}}{\left(1 + \frac{1}{x^3}\right)}$ $\frac{1}{x^5} \to 0$, $\frac{1}{x^3} \to 0$ $= -\sqrt{9} = -3$

HA as $x < 0$ is $y = -3$?? HA as $x > 0$ is $y = 3$

(3) $\lim_{x \to -\infty} (x^4 + x^5) = \lim_{x \to -\infty} x^4(1 + x) = \boxed{-}\infty$

Faktoriseer!

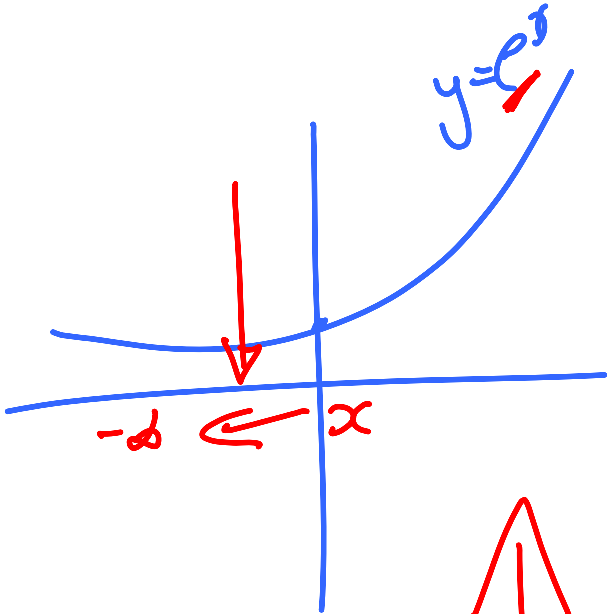
$+x$

? $\lim_{x \to \infty} (x^4 + x^5) = $?

(4) $\lim_{x \to \infty} \frac{1 - e^x}{1 + 2e^x}$

$$= \lim_{x \to \infty} \frac{\boxed{\frac{1}{e^x}} - \frac{\cancel{e^x}}{\cancel{e^x}}}{\frac{1}{e^x} + 2\frac{\cancel{e^x}}{\cancel{e^x}}} = \frac{0 - 1}{0 + 2} = -\frac{1}{2}$$

$\to 0$, $\to \infty$



as $x \to 0$ is HA $y = -\frac{1}{2}$

???? $\lim_{x \to -\infty} \frac{1 - e^x}{1 + 2e^x}$ $\therefore$ HA is $y = 1$

$e^x \to 0$